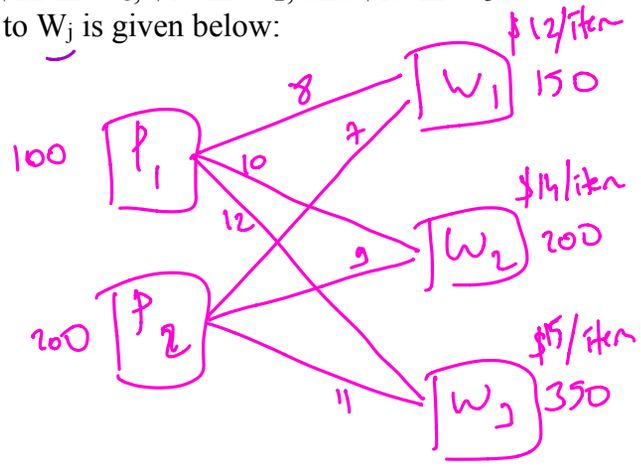
A Transportation Model: A company has two plants and three warehouses. Plant P_1 can produce at most 100 units and Plant P_2 can produce at most 200 units of the same product. The sales potential at Warehouse W_1 is 150, 200 at Warehouse W_2, and 350 at Warehouse W_3. The sales revenue per unit at the warehouses are: \$12 in W_1, \$14 in W_2, and \$15 in W_3. The cost of manufacturing one unit at P_i and shipping it to W_j is given below:

given →

P_i	W_1	W_2	W_3
1	8	10	12
2	7	9	11



Determine a transportation plans that maximizes overall profit.

Decision Variables: x_{ij} : # of units produced in plant i and sold in warehouse j

Objective Function: maximize profit = revenue − cost

$x_{11} + x_{21}$ → the amount sold in W_1 | $x_{11} + x_{12} + x_{13}$ → the amount produced in P_1

$x_{12} + x_{22}$ → " W_2 | $x_{21} + x_{22} + x_{23}$ → " P_2

$x_{13} + x_{23}$ → " W_3

Complete LP Model:

$$\text{maximum } z = 12(x_{11}+x_{21}) + 14(x_{12}+x_{22}) + 15(x_{13}+x_{23}) - 8x_{11} - 10x_{12} - 12x_{13} - 7x_{21} - 9x_{22} - 11x_{23}$$

s.t

$$\begin{aligned} x_{11} + x_{12} + x_{13} &\le 100 \\ x_{21} + x_{22} + x_{23} &\le 200 \\ x_{11} + x_{21} &\le 150 \\ x_{12} + x_{22} &\le 200 \\ x_{13} + x_{23} &\le 350 \end{aligned}$$

$x_{ij} \ge 0 \quad i = 1, 2, \quad j = 1, 2, 3.$

A Multi-Period Inventory Model: A producer of polyurethane has a stock orders for the next n periods (e.g., 6 months); let d_i be the known demand in period i. Let c_i be the cost of producing a gallon during period i and K_i be the maximum amount that can be produced in period i (because of capacity limitation). Finally, let h_i denote the per unit cost of inventory in stock at the end of period i. Suppose that I_0 be the initial inventory at the beginning of period 1. Find a production inventory-holding plan that will satisfy the known demand schedule at minimum total cost.

Decision variable: X_i : gallons produced in period i.

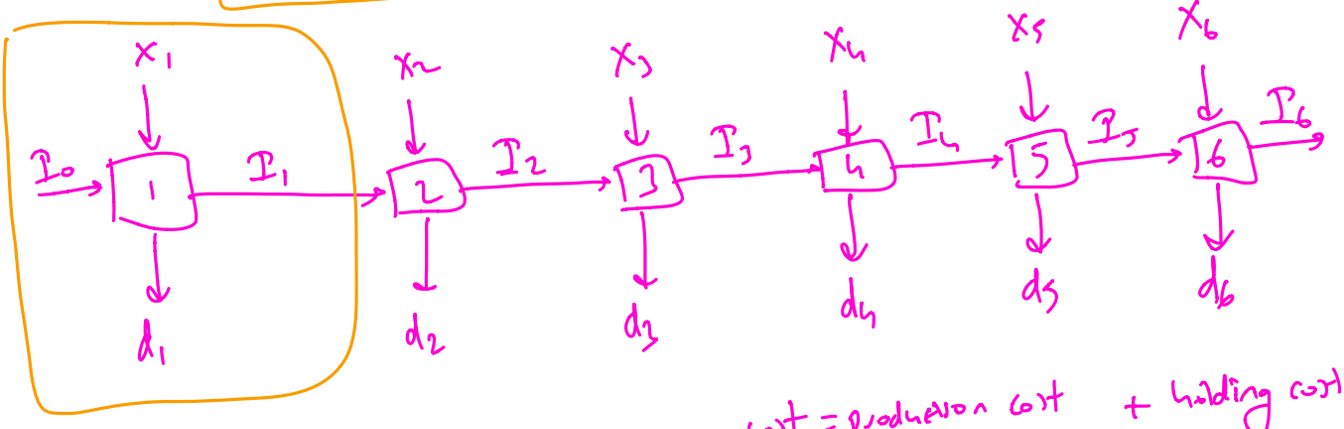
d_i
X_i
I_{i-1}
I_i

Define a state variable

I_i = gallons of inventory at the end of period i. So,

$I_{i-1} + X_i = d_i + I_i$

$I_{i-1} + X_i - d_i - I_i = 0$



cost = production cost + holding cost

Complete LP Model:

minimize z = production cost + holding cost = $\sum_{i=1}^{6} c_i \cdot X_i + \sum_{i=1}^{6} h_i \cdot I_i$

s.t

$I_0 + X_1 - I_1 - d_1 = 0$
$I_1 + X_2 - I_2 - d_2 = 0$
$I_2 + X_3 - I_3 - d_3 = 0$
$I_3 + X_4 - I_4 - d_4 = 0$
$I_4 + X_5 - I_5 - d_5 = 0$
$I_5 + X_6 - I_6 - d_6 = 0$

$X_i \le K_i$ $\quad i = 1, \dots, 6$ production capacity limitation
$I_i \ge 0$ $\quad i = 1, \dots, 6$
$X_i \ge 0$ $\quad i = 1, \dots, 6$

Investment (Loan Policy Model)

Thriftem Bank is in the process of devising a loan policy that involves a maximum of $12 million. The following table provides the pertinent data about available types of loans.

Type of Loan	Interest Rate	Bad-debt ratio
Personal	.140	.10
Car	.130	.07
Home	.120	.03
Farm	.125	.05
Commercial	.100	.02

90%

14% → 0.90 x_1

0.14 × 0.90 × x_1

Bad debts are unrecoverable and produce no interest value.

Competitions with other financial institutions require that the bank allocate at least 40% of the funds to farm and commercial loans. To assist the housing industry in the region, home loans must equal 50% of the personal, car and home loans. The bank also has stated policy of not allowing the overall ratio of bad debts on all loans to exceed 4%.
Thriftem Bank wants to maximize its net return, the difference between interest revenue and lost bad debts.

x_1 = personal loans (in millions of dollars)

x_2 = car loans

x_3 = home loans

x_4 = farm loans

x_5 = commercial loans.

Total interest = $0.14(0.90x_1) + 0.13(0.95x_2) + 0.12(0.97x_3)$
$+ 0.125(0.95x_4) + 0.100(0.98x_5)$

Bad debt = $0.10x_1 + 0.07x_2 + 0.03x_3 + 0.05x_4 + 0.02x_5$

Objective function

maximize z = total interest − bad debt

$= 0.126x_1 + 0.120x_2 + 0.1064x_3 + 0.11875x_4 + 0.098x_5$
$- (0.1x_1 + 0.07x_2 + 0.03x_3 + 0.05x_4 + 0.02x_5)$

(constraint) :

$$x_4 + x_5 \geq 0.4(x_1 + x_2 + x_3 + x_4 + x_5)$$

$$x_3 \geq 0.5(x_1 + x_2 + x_3)$$

$$0.1x_1 + 0.07x_2 + 0.03x_3 + 0.05x_4 + 0.02x_5 \leq 0.04(x_1 + x_2 + x_3 + x_4 + x_5)$$

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq 12$$

The complete model

Maximize $z = 0.026x_1 + 0.0509x_2 + 0.0864x_3 + 0.0875x_4 + 0.078x_5$

s.t

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq 12$$

$$0.4x_1 + 0.4x_2 + 0.4x_3 - 0.6x_4 - 0.6x_5 \leq 0$$

$$0.5x_1 + 0.5x_2 - 0.5x_3 \leq 0$$

$$0.06x_1 + 0.03x_2 - 0.01x_3 + 0.01x_4 - 0.02x_5 \leq 0$$

$$x_i \geq 0 \qquad i = 1, \ldots, 5$$

Single Period Production Model

In preparation for the winter season, a clothing company is manufacturing parka and goose overcoats, insulated pants and gloves. All products are manufactured in four departments: cutting, insulating, sewing and packaging. The company has received firm orders for its products. The contract stipulates a penalty for undelivered items. The following table provides the pertinent data of situation.

	Time per units (hr)				
Department	**Parka**	**Goose**	**Pants**	**Gloves**	**Capacity (hr)**
Cutting	.30	.30	.25	.15	1000
Insulating	.25	.35	.30	.10	1000
Sewing	.45	.50	.40	.22	1000
Packaging	.15	.15	.10	.05	1000
Demand	800	750	600	500	
Unit Profit	$30	$40	$20	$10	
Unit Penalty	$15	$20	$10	$8	

Devise an optimal production plan for the company.

Net receipts = Total profit − Total penalty

$30x_1 + 40x_2 + 20x_3 + 10x_4$

$15s_1 + 20s_2 + 10s_3 + 8s_4$.

x_1 = number of parka jackets

x_2 = " goose "

x_3 = " pairs of pants

x_4 = " " gloves

s_j = number of shortage units of product j, $j = 1, \ldots, 4$.

$x_1 + s_1 = 800$, $x_2 + s_2 = 750$, $x_3 + s_3 = 600$, $x_4 + s_4 = 500$

Objective function

maximize $z = 30x_1 + 40x_2 + 20x_3 + 10x_4 - (15s_1 + 20s_2 + 10s_3 + 8s_4)$

s.t

$0.30x_1 + 0.30x_2 + 0.25x_3 + 0.15x_4 \le 1000$ (cutting)

$0.25x_1 + 0.35x_2 + 0.30x_3 + 0.10x_4 \le 1000$ (insulating)

$0.45x_1 + 0.50x_2 + 0.40x_3 + 0.22x_4 \le 1000$ (sewing)

$0.15x_1 + 0.15x_2 + 0.10x_3 + 0.05x_4 \le 1000$ (packaging)

$x_1 + s_1 = 800$

$x_2 + s_2 = 750$

$x_3 + s_3 = 600$

$x_4 + s_4 = 500$

$x_j \geq 0, \; s_j \geq 0, \; j = 1, \ldots, 4.$

An Approach for Modeling

Step 1- Picture & Paraphrase: Initial understanding of the problem.

(i) Work in a flow-chart, input-output diagram, block diagram, process diagram, graph, schematic, etc. of the situation.

(ii) Then write a description of the problem without formula or data.

(iii) Show these to person(s) familiar with the problem for feedback & make sure that you are taking into account everything correctly.

Step 2 – Verbal Model: Break the problem into smaller elements.

(i) *Decisions* – something that can be evaluated or set to a value (how much to make, how many to hire/fire/promote, which warehouse to use, etc.)

(ii) *Constraints* – something that places limitation and/or requirement in the situation (budget, available rates of return, capacity, etc.). Constraints may fall in the following categories,
Inputs– what goes in the process? Are raw materials, capital, or labor restricted or limited in supply? Are there minimum amounts of input required?
Outputs– what is being produced? Are there sales requirements or limitations? Does a certain amount have to be kept in inventory?
Capacity– what capacity (time, labor, space) keeps you either from producing an infinite amount or from making more profit?
Ratio/Balance– what ratio between products or balance of ingredients must be maintained?

(iii) *Objective function* – what are you trying to optimize? May have many types of objective functions. How will it be measured (revenue, profit, market share, net present value, service level, etc.). Decision variables should directly or indirectly affect the objective. Should it be maximized or minimized?

Step 3 – Algebraic Model: transform verbal components of the model into algebraic representations.

Decisions– choose names or labels that describe each alternative, decide on their units.
Constraints– write each term as coefficient times the variables on the left-hand-side, an inequality (≤ or ≥) or equality (=) sign, and a constant on the right-hand-side. Three types of constraints: Restrictions (≤), Requirements (≥) or Equalities (=).
Objective function– write as coefficients times decision variables (to be maximized or minimized).

Step 4: Computer Model: transfer the algebraic model to a spreadsheet (e.g. Excel) or other optimization software (e.g. TORA). Follow the spreadsheet/software input format. Document the model with comments. Once model has been solved, the results may have to be put in a more readable form (with additional comments, graphs, charts and/or written report of what has been discovered); needs validity and may need to perform what-if or sensitivity analysis.